

Course Number	0418-346	Course Title	Declarative Programming
Credit Hours	3 (3:0:3)	Course Coordinator	Dr. Husain Ibraheem
Compulsory	No	Prerequisite	0418-201

Textbook

<i>Title:</i>	<i>ML for the Working Programmer</i>
<i>Author:</i>	<i>L.C. Paulson</i>
<i>Publisher:</i>	<i>Cambridge University Press, Latest Edition</i>
<i>Title:</i>	<i>PROLOG: Programming for Artificial Intelligence</i>
<i>Author:</i>	<i>I. Bratko</i>
<i>Publisher:</i>	<i>Addison-Wesley, Latest Edition</i>

Major Topics Covered in the Course

	Topics	Hours
1	Basic functional programming concepts.	7
2	Abstraction support in functional programming.	10
3	Extensible type system of SML.	3
4	Introduction to logic programming.	6
5	Programming in Prolog: Data objects: constants, variables, lists, structures; Clauses and evaluation: facts, rules, goals and queries, scope rules, substitution, matching vs. unification, most general unifier, execution of Prolog programs and derivation trees, the non-pure/non-declarative nature of Prolog programs (clause and goal orderings).	5
6	Advanced Logic Programming in Prolog: Controlling program execution: backtracking, the control operators fail and cut , negation by failure; Built-in predicates: arithmetic predicates (operators), structure predicates, system predicates (assert , retract , and consult); Optimization techniques: difference lists, tail recursion and the logical formulation of iterative definitions.	8
7	Exams.	3
	Total.	42

Grading

20%	Midterm 1
20%	Midterm 2
20%	Hmk/quiz/project
40%	Final

Course Outcomes

Students should be able to:

1. Understand the fundamental ideas underlying the functional and logic programming paradigms.
2. Write programs in declarative programming, in both of its functional and logic forms.
3. Think declaratively about computing solutions and to express their analysis in mathematical style.
4. Continue further studies and professional development in programming languages, artificial intelligence and expert systems.

Relationship between Course Outcomes and Student Outcomes

Course Outcomes	Student Outcomes					
	1	2	3	4	5	6
1	M	M				
2	H	H				H
3	H	H				
4				M		